

Fictitious Play for Continuous Games Revisited

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Abstract: In 1951 Julia Robinson proved that an iterative process for finding the value of a game, proposed by George W. Brown, converged for matrices. Later, using a proof closely modeled on Robinson's the author proved that the process converges for continuous games over compact sets. That proof was published in 1955. In this paper a generalized version is proved for the case when the successive maximizations and minimizations are inaccurate, and at the same time the presentation of 1955 is greatly improved.

1. Introduction

Sometime before 1951 [see *Robinson*] George W. Brown proposed an iterative method for finding the value of a matrix game. It was as follows. At stage 0 the two players choose a row i_0 and a column j_0 arbitrarily. At stage $N \geq 1$ suppose the choices $i_0, \dots, i_{N-1}$ and $j_0, \dots, j_{N-1}$ already made. Then choose i_N so as to maximize $\frac{1}{N} \sum_{n=0}^{N-1} a_{i_n j_n}$, and j_N so as to minimize $\frac{1}{N} \sum_{n=0}^{N-1} a_{i_n j_n}$. Suppose the corresponding maximum and minimum are $\bar{v}_N$ and $\underline{v}_N$ respectively. Evidently $\bar{v}_N \geq v \geq \underline{v}_N$, where v is the value of the game. It was Brown's conjecture that both $\bar{v}_N$ and $\underline{v}_N$ converged to v . The proof turned out to be very difficult; *Robinson* [1951] found it.

The author [1954] published a proof of the same convergence for continuous games over the product of two compact sets. In this paper I generalize that result to the case when the successive maximizations and minimizations are inaccurate, and, at the same time, substantially simplify and clarify the argument of 1955.

Because of its notoriously slow speed, the Brown-Robinson method is a method of last resort. The author originally had an application for these results. But after this paper was written he found a much better way to solve the game in question, and the application vanished. But there will always remain more games for which it is the only hope.

2. Description of the Iterative Process. The Convergence Theorem

We are given a function $g(u, v)$, continuous over the product of two compact spaces U and V . The u -player wishes to maximize $g(u, v)$, and the v -player to mini-

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mize it. We seek to approximate the value of the game by an iterative process, as follows.

At stage 0 the players choose elements $u_0 \in U$ and $v_0 \in V$ arbitrarily. At stage $N \geq 1$ suppose the choices $u_0, \dots, u_{N-1}$ and $v_0, \dots, v_{N-1}$ already made. The maximizer then chooses u_N so as to approximate the maximum M_N of the expectation $\frac{1}{N} \sum_{n=0}^{N-1} g(u, v_n)$ to accuracy no worse than ζ , where $\zeta \geq 0$, and the minimizer chooses v_N so as to approximate the minimum m_N of the expectation $\frac{1}{N} \sum_{n=0}^{N-1} g(u_n, v)$ to accuracy no worse than η , where $\eta \geq 0$.

The main result of this paper is the following.

Theorem. For the above iterative process,

$$\limsup_{N \rightarrow \infty} (M_N - m_N) \leq \zeta + \eta. \quad (2.1)$$

My main contribution to the proof of this theorem is the passage from the finite to the infinite, and the analysis of error. Otherwise the structure resembles the proof of Robinson [1951].

3. The Transposed Game

We shall need this important detail in section 7 below.

Consider the game defined by $g(u, v)$ but with the roles of the players reversed, i.e. the u -player seeks to minimize and the v -player to maximize. We denote the value of this "transposed game" by v^T . Now consider the expression

$$E_N(u) = \frac{1}{N} \sum_{n=0}^{N-1} g(u, v_n). \quad (3.1)$$

This may be regarded as the expectation to the u -player against the mixed strategy for the v -player consisting of playing each of the v_n , $n = 0, \dots, N-1$, with probability $1/N$. If this were by chance the v -player's optimal strategy, we would have $\text{Min } E_N(u) = v^T$. Since it is here an arbitrary maximizer strategy, we have in fact

$$\text{Min}_{u \in U} E_N(u) \leq v^T. \quad (3.2)$$

Similarly, for any set u_n , $n = 0, \dots, N-1$,

$$\text{Max}_{v \in V} \frac{1}{N} \sum_{n=0}^{N-1} g(u_n, v) \geq v^T. \quad (3.3)$$

Inequalities (3.2) and (3.3) are what we wanted for section 7; from now on the u -player is once again the maximizer and the v -player the minimizer.

4. The Generalized Formulation. The Lemma

We shall actually prove a statement somewhat stronger than the theorem of section 2, to which we shall henceforth refer as the *Convergence Theorem*. We introduce arbitrary continuous *initial functions* $\pi(u)$ and $\rho(v)$ on U and V respectively, and alter the iterative process as follows. At stage 0, u_0 is chosen so as to maximize $\pi(u)$ exactly, and v_0 so as to minimize $\rho(v)$ exactly. At stage $N \geq 1$, with the choices $u_0, \dots, u_{N-1}$ and $v_0, \dots, v_{N-1}$ already made, u_N is then chosen so as to maximize the function

$$F_N(u) = \pi(u) + \sum_{n=0}^{N-1} g(u, v_n) \quad (4.1)$$

to accuracy no worse than $N\xi$, and v_N so as to minimize the function

$$G_N(v) = \rho(v) + \sum_{n=0}^{N-1} g(u_n, v) \quad (4.2)$$

to accuracy no worse than $N\eta$.

Denote the exact maximum value of $F_N(u)$ by $\bar{u}_N$, and the exact minimum value of $G_N(v)$ by $\underline{v}_N$. Put

$$\bar{\mu} = \text{Max } \pi(u), \quad \underline{\mu} = \text{Min } \rho(v). \quad (4.3)$$

We shall prove the following lemma.

Lemma. For any $\epsilon > 0$ there exists an N_0 such that if $N \geq N_0$

$$\bar{\mu}_N - \underline{\mu}_N \leq (\xi + \eta + \epsilon)N + \bar{\mu} - \underline{\mu} \quad (4.4)$$

for any pair π, ρ of initial functions whatever.

That the Lemma implies the convergence theorem is obvious. We shall deduce the Lemma from a Proposition, stated in the following section 5, whose proof, by finite induction, takes up sections 6–9.

5. The Induction Hypothesis

We are going to prove the Lemma by induction relative to finite coverings of U and V by open sets, as follows.

Let $u \in U$. Because g is uniformly continuous on $U \times V$, there exists an open set $\mathcal{O}_u$ containing u such that

$$|g(u', v) - g(u, v)| \leq \epsilon/8 \quad (5.1)$$

for all $u' \in \mathcal{O}_u$ and $v \in V$. Because U is compact, there exists a finite subcollection $\{\mathcal{O}_{u_i}\}_{i=1, \dots, p}$ of the collection $\{\mathcal{O}_u\}_{u \in U}$ which covers U . If both u and u' are in some $\mathcal{O}_{u_i}$, evidently

$$|g(u, v) - g(u', v)| \leq \epsilon/4 \quad (5.2)$$

for all $v \in V$. Denote by K_i the closure of ∂_{u_i} . Evidently (5.2) continues to hold for pairs u, u' lying in K_i . We shall say that *the oscillation of g relative to u on K_i is not greater than $\epsilon/4$* .

Similarly we find a covering of V by open sets $\partial_{v_j}, j = 1, \dots, q$, having closures L_j , and such that the oscillation of g relative to v on L_j is not greater than $\epsilon/4$.

(I, J)-play. Denote by I_0 the whole set of indices $i = 1, \dots, p$, and by I any of its nonempty subsets. Similarly we write $J_0 = \{1, \dots, q\}$ and denote any of its nonempty subsets by J . The signs $\subseteq$ and $\subset$ denote inclusion and proper inclusion respectively. We write

$$U_I = \bigcup_{i \in I} K_i, \quad V_J = \bigcup_{j \in J} L_j, \quad (5.3)$$

so that $U = U_{I_0}$ and $V = V_{J_0}$. By *(I, J)-play* we mean the action of the iteration when u is restricted to U_I and v to V_J . We shall prove the following proposition, by finite induction.

Proposition. Consider any pair (I, J) and the corresponding (I, J)-play. To this play there corresponds an $\epsilon_{I,J}$ satisfying $\epsilon/2 \leq \epsilon_{I,J} < \epsilon$, and an $N_{I,J}$, each depending only on g, I and J , such that if $N \geq N_{I,J}$

$$\bar{\mu}_N - \underline{\mu}_N \leq (\xi + \eta + \epsilon_{I,J})N + \bar{\mu} - \underline{\mu}. \quad (5.4)$$

for any pair π, ρ of initial functions whatever.

The case $(I, J) = (I_0, J_0)$ obviously implies the Lemma and so the Theorem.

We first prove the Proposition for one-element sets. Suppose $I = \{i\}$ and $J = \{j\}$.

Put

$$\underline{v} = \max_{u \in K_i} \min_{v \in L_j} g(u, v), \quad \bar{v} = \min_{v \in L_j} \max_{u \in K_i} g(u, v). \quad (5.5)$$

Fix on a $u^* \in K_i$, and suppose v^* minimizes $g(u, v)$. Recall the definition at (4.1). We have

$$F_N(u^*) \leq \pi(u^*) + N \cdot g(u^*, v^*) + N\epsilon/4 \leq \pi(u^*) + N\underline{v} + N\epsilon/4. \quad (5.6)$$

Hence $\bar{\mu}_N \leq N\underline{v} + N\epsilon/4 + \bar{\mu}$. Similarly we get $\underline{\mu}_N \geq N\bar{v} - N\epsilon/4 + \underline{\mu}$. Since $\bar{v} \geq \underline{v}$ these yield the inequality

$$\bar{\mu}_N - \underline{\mu}_N \leq (\epsilon/2)N + \bar{\mu} - \underline{\mu}, \quad (5.7)$$

so that (5.4) holds with $\epsilon_{I,J} = \epsilon/2$.

For the next four sections we shall fix on a pair (I, J) , not both of which are one-element sets, and on the following hypothesis.

Induction Hypothesis. The Proposition is true for all pairs (I', J) with $I' \subset I$ and all pairs (I, J') with $J' \subset J$.

Our objective is to prove that the Proposition then holds for the pair (I, J) . This takes up sections 6–9.

6. General Plan of the Induction Step. Eligibility

Consider N stages of the (I, J) -play. We shall study these by breaking them into M groups of P stages each, with a “tail”, i.e. $N = MP + r$, where $0 \leq r < P$. P will be fixed, and M will vary with N . The induction hypothesis will then be applied, when it can be applied, to the individual groups.

The choice of P , $\epsilon_{I,J}$ and $N_{I,J}$. This is the logical point at which to make these choices, which will explain themselves in section 8 and at (9.4). We take ϵ^* to be the largest of the numbers $\epsilon_{I',J}$ for $I' \subset I$ and $\epsilon_{I,J'}$ for $J' \subset J$. Then $\epsilon/2 \leq \epsilon^* < \epsilon$. Next take P to be the largest of the integers $N_{I',J}$ for $I' \subset I$ and $N_{I,J'}$ for $J' \subset J$. Put

$$\epsilon_{I,J} = \frac{\epsilon + \epsilon^*}{2}. \quad (6.1)$$

Let

$$C = \max_{U \times V} |g(u, v)|. \quad (6.2)$$

Choose M^* so that

$$M^* > \frac{12C}{\epsilon - \epsilon^*}, \quad (6.3)$$

and then $N_{I,J} = M^*P$. If then $N \geq N_{I,J}$, $M \geq M^*$.

Eligibility. Consider one of the groups of P stages. The index $i \in I$ is said to be *eligible* on that group if, for some stage n^* in that group, there is a $u \in K_i$ which maximizes $F_N(u)$, as given by (4.1), to accuracy $n^*\zeta$. We define similarly eligibility of the minimizer indices j .

In what follows we distinguish two cases. In the first case, all of the $i \in I$ and $j \in J$ are eligible on the group. In this case each player is finding his expectation function roughly constant, which, if one ignores for the moment the initial functions, means that the other player is somewhere near an optimal mixed strategy. This in turn means that the quantities $\bar{\mu}_N$ and $\underline{\mu}_N$ ought not to be too far from each other. We estimate the difference $\bar{\mu}_N - \underline{\mu}_N$ for this case in the following section 7. If on the other hand some $i \in I$ or $j \in J$ is not eligible on the group, we can apply the induction hypothesis to the play across that group, with initial functions determined by the previous play, and obtain an estimate of the rise in $\bar{\mu}_N - \underline{\mu}_N$ across that group. This we do in section 8. And in section 9 we put the results together to achieve the induction step.

7. Study of a Group: All $i \in I$ and $j \in J$ Eligible

Fix attention on a group, indexed by $m \geq 1$, for which all the indices $i \in I$ and $j \in J$ are eligible. Our objective is to estimate $\bar{\mu}_{mP} - \underline{\mu}_{mP}$; that estimate is given at (7.10).

We do this by relating $\bar{\mu}_{mP}$ and $\underline{\mu}_{mP}$ to the value v^T of the transposed game of section 3.

By (3.2) applied to the transposed game over $U_I \times V_J$, there exists a $u^* \in U_I$ such that

$$E_{mP}(u^*) \leq v^T. \quad (7.1)$$

Now $u^* \in K_i$ for some $i \in I$; and since i is eligible there exists a stage n^* on the group, i.e. with $(m-1)P \leq n^* \leq mP-1$, and a $u^{**} \in K_i$ for which

$$\pi(u^{**}) + n^*E_{n^*}(u^{**}) \geq \bar{\mu}_{n^*} - n^*\zeta. \quad (7.2)$$

Because both u^* and u^{**} are in K_i ,

$$\begin{aligned} n^*E_{n^*}(u^{**}) &\leq mPE_{mP}(u^{**}) + PC \leq mPE_{mP}(u^*) + mP\epsilon/4 + PC \leq \\ &\leq mPv^T + mP\epsilon/4 + PC. \end{aligned} \quad (7.3)$$

On combining this with (7.2) we get

$$\bar{\mu}_{n^*} \leq mP(v^T + \zeta + \epsilon/4) + \bar{\mu} + PC, \quad (7.4)$$

where

$$\bar{\mu} = \max_{u \in U_I} \pi(u). \quad (7.5)$$

Now suppose u^{***} maximizes $F_{mP}(u)$. Then

$$\bar{\mu}_{mP} = F_{mP}(u^{***}) \leq F_{n^*}(u^{***}) + PC \leq \bar{\mu}_{n^*} + PC. \quad (7.6)$$

On combining this with (7.4) we get the desired estimate for $\bar{\mu}_{mP}$:

$$\bar{\mu}_{mP} \leq mP(v^T + \zeta + \epsilon/4) + \bar{\mu} + 2PC. \quad (7.7)$$

By the mirror argument, using (3.3), we get

$$\underline{\mu}_{mP} \geq mP(v^T - \eta - \epsilon/4) + \underline{\mu} - 2PC, \quad (7.8)$$

where

$$\underline{\mu} = \min_{v \in V_J} \rho(v). \quad (7.9)$$

On combining (7.7) and (7.9) we achieve the desired estimate of $\bar{\mu}_{mP} - \underline{\mu}_{mP}$:

$$\bar{\mu}_{mP} - \underline{\mu}_{mP} \leq mP(\zeta + \eta + \epsilon/2) + \bar{\mu} - \underline{\mu} + 4PC. \quad (7.10)$$

8. Study of a Group: Some i or j Ineligible

Suppose some i' is ineligible over a group with index $m \geq 2$. Put $I' = I - \{i\}$. Evidently I' is not empty, and it is a proper subset of I . By the induction hypothesis, the Proposition of section 5 holds for the pair (I', J) . Now the play of the long game, as it

is seen on the m -th group, may be regarded as the play of a P -stage game with the initial functions

$$\pi'(u) = \pi(u) + \sum_{n=0}^{(m-1)P-1} g(u, v_n) \quad (8.1)$$

and

$$\rho'(v) = \rho(v) + \sum_{n=0}^{(m-1)P-1} g(u_n, v), \quad (8.2)$$

reflecting the effect of the $(m-1)P$ previous choices. It was for just this step that the initial functions were introduced.

Now P was chosen in section 6 so that $P \geq N_{I',J}$, and ϵ^* so that $\epsilon_{I',J} \leq \epsilon^*$. Since the pair $N_{I',J}, \epsilon_{I',J}$ works for *any* pair of initial functions²), the Proposition for (I', J) implies that

$$\bar{\mu}_{mP} - \underline{\mu}_{mP} \leq (\zeta + \eta + \epsilon^*)P + \bar{\mu}' - \underline{\mu}', \quad (8.3)$$

where

$$\bar{\mu}' = \max_{u \in U_{I'}} \pi'(u), \quad \underline{\mu}' = \min_{v \in V_J} \rho'(v). \quad (8.4)$$

Now

$$\bar{\mu}' \leq \max_{u \in U_I} \pi'(u) = \bar{\mu}_{(m-1)P} \quad (8.5)$$

and $\underline{\mu}' \geq \underline{\mu}_{(m-1)P}$. Hence (8.3) may be continued to

$$\bar{\mu}_{mP} - \underline{\mu}_{mP} \leq \bar{\mu}_{(m-1)P} - \underline{\mu}_{(m-1)P} + (\zeta + \eta + \epsilon^*)P. \quad (8.6)$$

If $m = 1$ we have directly

$$\bar{\mu}_P - \underline{\mu}_P \leq \bar{\mu} - \underline{\mu} + (\zeta + \eta + \epsilon^*)P, \quad (8.7)$$

$\bar{\mu}$ and $\underline{\mu}$ being given by (7.5) and (7.9) respectively.

9. Completion of the Induction Step

Recall the definition of M^* at (6.4) and the definition of $N_{I,J}$ following it. We fix on an $N \geq N_{I,J}$. Then $N = MP + r$ with $M \geq M^*$ and $0 \leq r < P$.

Suppose first that there is an ineligible index on every group. Then we find immediately from the inequality (8.7) and the inequalities (8.6) that

$$\bar{\mu}_{MP} - \underline{\mu}_{MP} \leq (\zeta + \eta + \epsilon^*)MP + \bar{\mu} - \underline{\mu}, \quad (9.1)$$

$\bar{\mu}$ being given by (7.5) and $\underline{\mu}$ by (7.9). Next suppose there is a group with all the $i \in I$

²) This is perhaps the most delicate point in the whole argument.

and $j \in J$ eligible. Let m^* be the index of the last such group. If $m^* = M$, we have from (7.10) that

$$\bar{\mu}_{MP} - \underline{\mu}_{MP} \leq MP (\xi + \eta + \epsilon^*) + \bar{\mu} - \underline{\mu} + 4PC. \quad (9.2)$$

Suppose $m^* < M$. Then on combining the inequality (7.10) for $m = m^*$ with the inequalities (8.6) for $m = m^* + 1, \dots, M$, we get (9.2) once again. Hence (9.2) holds in any case.

We may estimate the tail crudely by the moduli. This, along with (9.2), gives us

$$\bar{\mu}_N - \underline{\mu}_N \leq MP (\xi + \eta + \epsilon^*) + \bar{\mu} - \underline{\mu} + 6PC. \quad (9.3)$$

Now from (6.3) and (6.1)

$$M\epsilon^* + 6C \leq M \left(\frac{\epsilon + \epsilon^*}{2} \right) + M\epsilon_{I,J}. \quad (9.4)$$

Hence (9.3) continues to

$$\bar{\mu}_N - \underline{\mu}_N \leq MP (\xi + \eta + \epsilon_{I,J}) + \bar{\mu} - \underline{\mu}. \quad (9.5)$$

The Proposition (see (5.4)) therefore holds for the pair (I, J) and the induction is accomplished. Hence the Lemma of section 4, and so the Theorem of section 2 are proved. $\square$

References

- Robinson, J.*: An iterative method of solving a game. *Annals of Mathematics* 2, 1951, 296–301.
Danskin, J.: Fictitious play for continuous games. *Naval Research Logistics Quarterly* 1, 1954, 313–320.

Received October 1979
 (revised version April 1981)